

Software Engineering Lab Specification Report

Incident Escalation & On-Call Rotation Engine

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System: DevOps Incident Escalation Engine	GitHub Repository: github.com/arjunsrikanthh/PES1UG24AM920_se_lab

1. System Overview

A DevOps on-call management system organizing team shift rotations, routing inbound monitoring alerts through multi-tiered phone/SMS/webhook escalation ladders, and tracking incident post-mortems to ensure high availability and minimize MTTR.

2. Requirements Specification

2.1 Functional Requirements

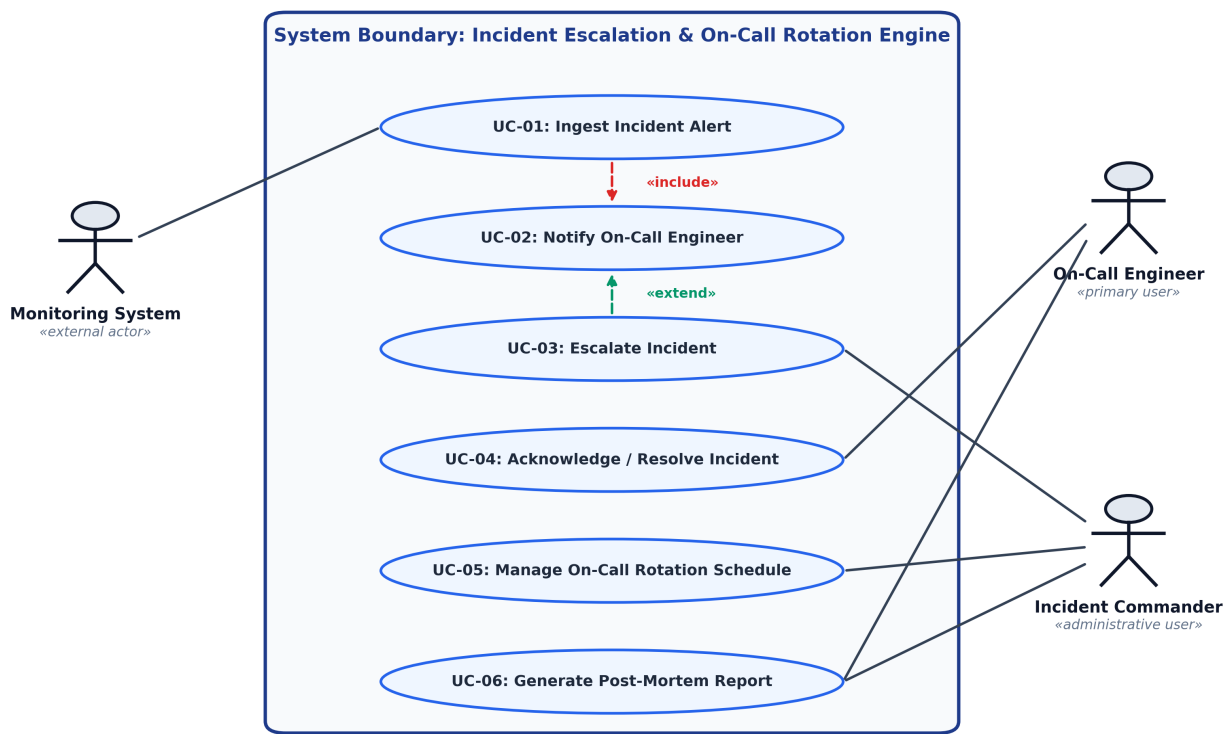
Req ID	Description	Pri.	Acceptance Criteria	Rationale
FR-001	Ingest incident alerts, identify active on-call engineer from rotation schedule, and escalate to secondary engineer if unacknowledged within 5 minutes.	High	Pass: Escalation triggered upon timeout. Fail: Unacknowledged P1 alert drops without escalation.	Core alerting ensures critical incidents are never unattended, reducing MTTR.
FR-002	Allow Incident Commander to define/modify multi-tiered escalation policies specifying notification channels (phone, SMS, email, webhook) and timeouts per tier.	High	Pass: Updated policy active in next cycle. Fail: Old policy persists after save.	Customizable ladders adapt workflows to varying team sizes, SLAs, and severity levels.
FR-003	Enable on-call engineers to acknowledge, reassign, or resolve incidents through web dashboard, SMS reply, or phone keypress.	High	Pass: Status update reflected within 10s. Fail: Status remains 'Open' after ACK.	Multi-channel ACK enables rapid triage away from workstations, slashing MTTR.
FR-004	Support creation and management of recurring on-call rotation schedules with configurable shift durations, manual overrides, and holiday substitutions.	Med	Pass: Correct engineer identified for queried timestamp, including overrides. Fail: Wrong engineer assigned.	Automated rotations eliminate scheduling errors and ensure 24/7 continuous coverage.
FR-005	Generate a post-mortem report template upon incident resolution, capturing event timeline, root-cause fields, and action items, linked to incident record.	Med	Pass: Template auto-populated with timeline and linked. Fail: Template missing or unlinked.	Structured post-mortems drive root-cause analysis and prevent repeat incidents.

2.2 Non-Functional Requirements

Req ID	Category	Description	Pri.	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Alert dispatch via webhook, SMS, and email must initiate within 3 seconds of alert ingestion.	High	Pass: Latency <= 3s under peak load (>= 500 concurrent alerts). Fail: Latency > 3s or unencrypted dispatch.	Rapid dispatch is vital for SLA compliance during P1/P2 incidents.
NFR-002	Availability & Security	Maintain 99.9% uptime with automatic failover; all data encrypted with TLS 1.2+ (in transit) and AES-256 (at rest).	High	Pass: 30-day uptime >= 99.9%; penetration test confirms no plaintext paths. Fail: Downtime > 0.1% or unencrypted paths.	System must remain operational during outages; encryption secures sensitive data.

3. UML Use-Case Model & Diagram

The use-case model defines the behavioral boundary of the Incident Escalation & On-Call Rotation Engine. It illustrates the interactions between automated monitoring systems, responding engineers, and operations leadership.



Element / UC	Type & Name	Behavioral & Architectural Description
Monitoring System	External Actor	Automated observability platforms (Prometheus, CloudWatch, Datadog) emitting alerts via authenticated webhooks.
On-Call Engineer	User Actor	Active primary/secondary engineer who triages, acknowledges, resolves incidents, and compiles post-mortems.
Incident Commander	Admin Actor	Engineering manager/lead who configures rotation schedules, manages escalation ladders, and handles exhausted tiers.
UC-01 & UC-02	«include» Dependency	<i>Ingest Incident Alert</i> unconditionally includes <i>Notify On-Call Engineer</i> upon successful ingestion and validation.
UC-03 & UC-02	«extend» Dependency	<i>Escalate Incident</i> conditionally extends <i>Notify On-Call Engineer</i> when Tier-1 SLA timeout (5 min) expires unacknowledged.
UC-04, 05, 06	Core Operations	<i>UC-04</i> : Multi-channel ACK/Resolve. <i>UC-05</i> : Rotation schedule & override setup. <i>UC-06</i> : Post-mortem report generation.

4. Detailed Use-Case Flow Specification: UC-01

4.1 Use Case Metadata & Execution Context

Use Case ID	UC-01	Use Case Name	Ingest Incident Alert & Notify On-Call Engineer
Primary Actor	Monitoring System (External)	Secondary Actors	On-Call Engineer, Incident Commander
Description	The system receives an incident alert from an external monitoring tool, identifies the currently active on-call engineer from the rotation schedule, and dispatches notifications through configured escalation channels. If unacknowledged within the configured timeout (5 min), it escalates to the next tier.		
Preconditions		Postconditions	
1. At least one on-call rotation schedule is active with assigned engineers. 2. Escalation policy with at least one tier configured for service/team. 3. Monitoring system integration (webhook endpoint) is active & authenticated.		1. Incident recorded with unique ID, severity (P1–P4), and timestamp. 2. On-call engineer notified via configured channels (phone/SMS/email/webhook). 3. Incident status updated to 'Acknowledged' or 'Escalated'.	

4.2 Main Success Scenario (Normal Flow)

Step	Actor	Action / System Response
1	Monitoring System	Sends incident alert payload (severity, service name, description) to webhook endpoint.
2	System	Validates alert payload format and authenticates cryptographic signature.
3	System	Creates new incident record with unique ID, timestamps event, and classifies severity (P1–P4).
4	System	Queries active rotation schedule to identify Tier-1 on-call engineer for affected service.
5	System	Retrieves escalation policy and determines notification channels and timeout for Tier-1.
6	System	Dispatches notifications to Tier-1 engineer via configured channels (phone, SMS, email) within 3s.
7	On-Call Engineer	Receives notification and acknowledges incident (via dashboard, SMS 'ACK', or phone keypress '1').
8	System	Records acknowledgement timestamp, updates status to 'Acknowledged', and halts escalation timer.
9	System	Displays incident on On-Call Engineer's dashboard with full context for investigation.
10	—	Use case ends successfully.

4.3 Alternate Flow: Acknowledgement Timeout & Multi-Tier Escalation

Step	Actor	Action / Exception Handling Procedure
7a	System	Tier-1 escalation timeout (5 minutes) expires without acknowledgement.
7a.1	System	Marks incident status as 'Escalated – Tier 2' and logs timeout telemetry event.
7a.2	System	Queries escalation policy for Tier-2 and identifies secondary on-call engineer.
7a.3	System	Dispatches notifications to Tier-2 engineer via configured secondary channels.
7a.4	System	Simultaneously alerts Incident Commander that a Tier-1 escalation timeout occurred.
7a.5	On-Call Engineer (T2)	Receives notification and acknowledges incident.
7a.6	System	Records ACK, updates status to 'Acknowledged (Tier-2)', stops escalation. Resumes at Step 9.
7a.5a	System	Exhaustion: If Tier-2 also times out, system alerts Incident Commander directly with 'Critical – All Tiers Exhausted' for manual intervention.